Swaminarayan Sampradaya library chat

Installation, user and admin guide, runbook, troubleshooting, monitoring
Version 2.12.0 · macOS · Windows · Ubuntu

Contents

1. Installation
2. Reader guide
3. Admin guide
4. Runbook
5. Monitoring and service tracking
6. Troubleshooting
7. Configuration
8. Browsers, phones and tablets

1. Installation

The Swaminarayan Sampradaya library chat (internal name **Granth**) runs on one computer and is used from a web browser on that computer. It is free to run: books stay on the computer, and the AI that writes answers is either free (Ollama on the computer, or a free online key) or optional.

	macOS	Windows	Ubuntu / Debian
Package	`granth-mac-<version>.zip`	`granth-windows-<version>.zip`	`granth-ubuntu-<version>.zip`
Installer	`granth_setup.sh`	`granth_setup.cmd` (runs `granth_setup.ps1`)	`granth_setup.sh`
Supported	macOS 12 or newer, Apple Silicon or Intel	Windows 10 (1809+) and 11	Ubuntu 22.04 / 24.04, Debian 12
Installs to	`~/granth`	`%LOCALAPPDATA%\Granth`	`~/granth`
Books folder	`~/GranthBooks`	`%USERPROFILE%\GranthBooks`	`~/GranthBooks`
Runs as	launchd agent `com.granth.app`	scheduled task "Granth library"	systemd user service `granth.service`

What you need

- **Memory:** 4 GB minimum; 8 GB recommended. 16 GB or more if you use Ollama (AI on the computer) or the offline Indian-language translator.
- **Disk:** about 3 GB for the app with meaning-based search, plus your books, plus about 5 GB for Ollama models or 4 GB for the offline translator if you use them.
- **Internet:** for installation, and for the online AI if you choose that. Readers use the library through a browser on the same computer.
- **An AI for writing answers** (choose during setup, change later with the `ai` command):
- **Free online key** (recommended for Gujarati and Hindi: Google Gemini, https://aistudio.google.com). Groq, Cerebras, NVIDIA, OpenRouter and Mistral also have free keys.
- **Ollama**: runs on the computer, no key, works offline, needs 8 GB+ memory.
- **Skip**: readers then see the matching passages with page and line numbers, without a written answer.

macOS

1. Install Homebrew if you do not have it: https://brew.sh
2. Unzip `granth-mac-<version>.zip`, open **Terminal** in that folder and run:

   ```
   chmod +x granth_setup.sh
   ./granth_setup.sh
   ```

3. Setup installs Python 3.12, Tesseract (with Indian languages) and Poppler with Homebrew, creates the app's Python environment, asks which AI to use, and starts the background service.

It opens the library in your browser when it is ready and prints the **admin token**.
4. Put books in `~/GranthBooks`. Keep this folder **outside** Documents, Desktop and Downloads: macOS does not let background apps read those folders.

Windows 10 and 11

1. Unzip `granth-windows-<version>.zip` (right-click > Extract All). Do not run it from inside the zip.
2. Double-click **`granth_setup.cmd`**. If Windows SmartScreen warns, choose *More info > Run anyway*.
3. Setup uses **winget** (built into Windows 10 1809+ and 11) to install, for your user only where possible:
   - Python 3.12, if no Python 3.10+ is found;
   - Tesseract OCR and Poppler (Windows may ask for permission), and downloads the Indian-language OCR data into the app's data folder.
4. It creates the app's Python environment in `%LOCALAPPDATA%\Granth`, asks which AI to use, creates the scheduled task **"Granth library"** (starts at log-on, restarts if it stops; if a task cannot be created it uses a Startup-folder shortcut instead), starts it and opens the library.
5. Put books in `%USERPROFILE%\GranthBooks`.

Later commands are run from `%LOCALAPPDATA%\Granth\granth_setup.cmd`, for example `granth_setup.cmd status` in a Command Prompt.

No winget? Install Python 3.12 from https://www.python.org/downloads/windows/ (tick *Add python.exe to PATH*) and Tesseract from https://github.com/UB-Mannheim/tesseract/wiki, then run setup again.

Ubuntu / Debian

1. Unzip `granth-ubuntu-<version>.zip` and in a terminal in that folder run:

   ```
   chmod +x granth_setup.sh
   ./granth_setup.sh
   ```

2. Setup asks for your password once (sudo) to install `python3-venv`, `poppler-utils` and `tesseract-ocr` with the Indian-language packs through apt. It then works as your user in `~/granth`.

3. It installs a **systemd user service** (`~/.config/systemd/user/granth.service`) and enables *lingering* so the library starts at boot even before you log in. Where there is no systemd user session (some containers and WSL), it runs as a background process and adds an `@reboot` entry to your crontab.
4. Put books in `~/GranthBooks`.

Run setup as your normal user, not with `sudo`.

Options for setup

Set these before running setup (macOS/Linux: `PORT=8080 ./granth_setup.sh`; Windows Command Prompt: `set PORT=8080` then `granth_setup.cmd`):

Option	Default	Meaning
`AI`	ask	`ollama`, `key` or `skip`
`BOOKS_DIR`	`~/GranthBooks`	folder that is watched for new books
`PORT` / `ADMIN_PORT`	`8000` / `8001`	library chat / admin page ports
`WITH_SEMANTIC`	`1`	`0` skips meaning-based search (saves ~1.5 GB; keyword search still works)
`WITH_INDIC`	`0`	`1` also installs the offline Indian-language translator
`OLLAMA_MODEL`	by memory	e.g. `qwen2.5:7b`
`PYTHON_BIN`	found automatically	a specific Python 3.10+
`SKIP_BACKUP`	`0`	`1` skips the database copy made before an update

After installation

- Library chat: http://localhost:8000. Admin page: http://localhost:8001, **this computer only**. Unlock it with the admin token (shown at the end of setup, and stored in the data folder as `admin_token`).
- Add books: copy them into the books folder (read automatically within 30 seconds), or upload them on the admin page (Books tab).
- Check everything: `granth_setup status` (see MONITORING.md).

Optional: offline Indian-language translation (IndicTrans2)

`granth_setup indic` installs AI4Bharat's IndicTrans2 in its own Python environment and downloads its models (~3 GB). The models are shared only with signed-in Hugging Face users who accept AI4Bharat's conditions (free): setup lists the three model pages, asks for a Hugging Face **Read** token, checks access, and only then downloads. Remove it again with `granth_setup indic remove`. A free Gemini key gives good Gujarati translation without it.

Updating

Download the new package and run its installer the same way. Settings, keys, chats, index and the audit trail are kept; the library database is copied to `data/backups` first, and the previous version is kept for `granth_setup rollback`. `granth_setup version` shows the installed version and the version in the package.

Uninstalling

Command	Removes
`granth_setup indic remove`	only the offline translator, its environment and models
`granth_setup uninstall`	the app, its environments and the service; asks before deleting chats, index, models and logs, Ollama, the OCR tools, and pip's download cache

Command	Removes
`granth_setup uninstall --all`	all of the above without asking

Your books folder is never deleted. Python itself is kept, because other programs may use it.

(`granth_setup` means `./granth_setup.sh` on macOS/Ubuntu, or `granth_setup.cmd` on Windows.)

Reaching it from other devices (phones, tablets) — optional

By default only the computer itself can open the library. To let phones and tablets on the same network use it, a reverse proxy with HTTPS on the computer is recommended (Caddy, nginx). The quick local-network way is `HOST=0.0.0.0` and `ALLOW_LAN=1` in the settings file, then restart; set an `ACCESS_CODE` so only people who know the code can chat. The admin page always stays on this computer only. See CONFIGURATION.md.

2. Reader guide: what the library chat does

Open the library in a browser (http://localhost:8000 on the library computer).

Asking

- Type or speak a question in any Indian language or English, and press **Enter** (or the arrow button). **Shift+Enter** starts a new line.
- The answer comes **from the books in the library**: it cites the passages it uses, like [1], and lists them under **References** with the book, page and line (or verse) so you can look them up. A green *Checked against the books* badge means every quoted sentence was found in the cited passage.
- If the books do not contain the answer you see *Not found in the books*, with the closest passages found. The question goes to the library admin, who can add an answer.
- Spelling mistakes are corrected using words that appear in the books; the answer says what was searched.
- Ask about the library itself: *"which books are in the library"*, *"summary of each book"*, *"summary of Vachanamrut"*.

Choosing books

All books (below the question box) opens the **Library**: tick books to search only those. With nothing ticked every book is searched. A question that mentions "all books" or "the library" searches the whole library and says so.

Language

Auto answers in the language of the question; choose a language to always answer in it. The menu also switches the screen language (English, ગુજરાતી, हिन्दी).

Under each answer

Button	What it does
Read aloud	reads the answer with a **male voice** (never a female voice)
Copy	copies the answer
Translate	translates the answer into another language, shown below the answer
Helpful / Not helpful	rates the answer; *Not helpful* lets you write what the answer should have been (the admin reviews it)

Each reference can be opened to read the passage, read it aloud, translate it, copy its reference, or **Show surrounding text** with line numbers.

Translation

- The first time you translate, the **terms** appear: translations are made by AI and can contain mistakes; the original book is the authority. Tick *I understand* and accept. If the admin changes

the terms you are asked again.
- Translations are labelled **AI translation**. If one is wrong, **Suggest a better translation** sends your correction to the admin, who can approve it so it is used from then on.
- While a translation runs, the question box and *Read aloud* are paused; **Stop** (the square button) cancels it.

Voice

- **Microphone** button (or **Alt+M**): speak your question. In browsers without voice input (Firefox) the computer's own speech recognition is used if installed; otherwise type.
- **Hands-free** (top right, or **Alt+V**): speak, hear the answer, speak again. It keeps listening through short silences, carries on after an answer that failed, and listening starts again when you press **Stop** while an answer is read. Say **"stop"** (or बंद / બંધ) to end it; after about a minute of silence it pauses by itself. Some browsers (Safari) need a tap on the microphone to listen again after the screen locks.
- **Stop** (square button, or **Alt+S**) stops reading aloud. **Alt+R** reads the last answer again.
- Settings: speaking speed, read answers automatically, also read source names.
- On phones and tablets sound plays only after a tap: tap **Read aloud**.

Downloads

Books the admin allows show **Download PDF** in the Library. The PDF has a cover page with the library's signature and a copy code on every page. Published translations show *" translation (AI), PDF"*; the terms are shown first and printed on its cover.

Chats

Chats are listed on the left (on phones: the sidebar button at the top left), grouped by day. Each can be renamed, downloaded or deleted. **New chat** starts a fresh one.

Display and accessibility

- Settings: light, dark or high-contrast theme; text size.
- Everything works with the keyboard and with screen readers (VoiceOver, NVDA, TalkBack): labelled buttons, a skip link, and spoken announcements of answers and status.
- Keyboard shortcuts: **Alt+M** microphone, **Alt+S** stop, **Alt+R** read again, **Alt+V** hands-free, **Alt+N** new chat, **Alt+/** question box. The **Keyboard shortcuts** item lists them.

3. Admin guide

The admin page runs on its own port (http://localhost:8001) and only answers on the library computer. Unlock it with the **admin token** (printed at the end of setup; stored as `admin_token` in the data folder). **Lock** forgets the token in this browser tab. Every admin action is in the audit trail.

Books

- **Upload** PDF, EPUB, Word (.docx), text, Markdown, web pages, and scans/photos (OCR in Indian languages), or copy files into the books folder (read within 30 seconds; **Scan books folder now** checks immediately).
- Status shows *queued*, *processing n%*, *ready* or *failed* (with the reason). **Read again** re-reads a book; **Remove** takes it out of the library (the file in the books folder is kept).
- **Download: Allow** lets readers download that book as a signed PDF (off by default).
- **Translate…** opens the Translations tab for that book.

Book sources

- **Folders on this computer**: the main books folder plus any folders you add. With *Google Drive for desktop* installed, add your Drive books folder (Mac: `~/Library/CloudStorage/GoogleDrive-<you>/My Drive/Books` with *Mirror files* chosen; Windows: `G:\My Drive\Books`). Removing a folder keeps its books in the library.
- **Google Drive folders (online)**, for any computer including Ubuntu servers:
- In console.cloud.google.com create a project and enable the **Google Drive API**.
- Create a **service account** (no roles needed), then *Keys > Add key > JSON*; upload the file here.
- In Drive, **share** each books folder with the service account's e-mail address as **Viewer**.
- Paste the folder link and choose **Add Drive folder**.

Sub-folders and folder shortcuts are included; PDF, EPUB, Word, text, web pages, images and Google Docs (exported as Word) are read. Each folder is checked again every 30 minutes (`DRIVE_SYNC_MINUTES`) or with **Check now**; only new or changed files are downloaded. Files removed from Drive are reported and stay in the library. The library only reads Drive. For folders shared as *Anyone with the link*, an **API key** is enough instead of a service account.

Unanswered

Questions the books could not answer, with the reason. **Teach answer** saves an approved answer that is given for that question from then on; **Mark resolved** clears it. Download as CSV.

Reader corrections

Corrections readers wrote after *Not helpful*. **Approve** makes it an approved answer; **Reject** discards it.

Translations

- **Translate a book**: choose the book and language. It runs in the background (pause, resume, cancel; continues after a restart; waits by itself when the free AI is busy).
- **Review**: original and translation side by side, 50 passages at a time. **Save correction** stores your wording and remembers it for future translations. Passages marked *needs checking* failed the automatic quality check; a book cannot be published until they are corrected.
- **Preview PDF**, **Publish** (readers can download it), **Unpublish**, **Add to library for questions** (the translation becomes a searchable book).
- **Reader suggestions**: edit if needed, then approve (remembered) or reject.
- **Terminology**: names and terms that must always be translated the same way. Sampradaya names (Shriji Maharaj, Vachanamrut, Vadtal, Gadhada, Akshardham…) are built in for Gujarati and Hindi.
- The line at the top says which engine translates Gujarati (AI provider or the offline IndicTrans2) and shows any problem with it.

Downloads & terms

- **Signature**: name, title, organisation, note, optional signature image; **Download a sample PDF**.
- **Terms and conditions for AI translation** in English, Gujarati and Hindi. Saving raises the version and every reader accepts again.

Learning

How the library learns without retraining the AI: ratings re-rank passages for similar questions, rephrased questions teach new words (approve or reject), and answers several readers found helpful can become approved answers. **Download approved answers (JSONL)**.

Visitors

Activity grouped by network address (today, 7, 30, 90 days, 1 year): last seen, actions, questions, translations, downloads, readers, busiest minute, and **flags** for unusual use (many requests in a minute, slowed down by the rate limit, refused access, many errors).

- Click an address: activity by day, what it did, browsers, latest records; download its records or open them in the audit trail.
- **Label** an address ("Temple office"); **Block…** for 1 hour, 24 hours, 7 days or until unblocked (this computer cannot be blocked). One address can be a whole office or mobile network.

Audit trail

Every question, answer, translation, download, correction and admin action, with time, address and browser. Search by address, action (or a group such as `translate…`), word, date range, or errors only; **Load older records**; **Download these records (CSV)**. **Check it has not been changed** verifies the tamper-evident chain (each record carries the fingerprint of the one before it).

Log files

Sizes of all logs, audit record count and oldest record, download any log, recent warnings and errors, and **Apply the retention period now**.

System

Name and symbol shown to readers (default *Swaminarayan Sampradaya* and ॐ): change them here; they appear in the page title and header, beside every answer, in downloaded chats and on PDFs. Readers see the change after reloading. **Back to the default** returns to `APP_NAME` / `APP_MARK` from the settings file.

Version, books, passages, search index, AI providers and their daily use, background load (worker threads, AI requests running and waiting), and audit settings.

4. Runbook

Day-to-day operation of the library computer. `granth_setup` below means `~/granth/granth_setup.sh` on macOS and Ubuntu, or `%LOCALAPPDATA%\Granth\granth_setup.cmd` on Windows.

Where things are

	macOS / Ubuntu	Windows
App	`~/granth/app`	`%LOCALAPPDATA%\Granth\app`
Settings and keys	`~/granth/app/.env`	`%LOCALAPPDATA%\Granth\app\.env`
Data (database, index, audio cache, models)	`~/granth/data`	`%LOCALAPPDATA%\Granth\data`
Logs	`~/granth/data/logs`	`%LOCALAPPDATA%\Granth\data\logs`
Database backups (made before each update)	`~/granth/data/backups`	`...\data\backups`
Admin token	`~/granth/data/admin_token`	`...\data\admin_token`
Books folder	`~/GranthBooks`	`%USERPROFILE%\GranthBooks`
Service	macOS: launchd `com.granth.app`; Ubuntu: `systemctl --user ... granth`	Task Scheduler: "Granth library"

Commands

Command	Use
`granth_setup status`	service state, both ports, Ollama, disk, error count
`granth_setup health`	`OK` / `FAIL` with exit code 0/1, for monitors
`granth_setup start` / `stop` / `restart`	control the background service
`granth_setup logs`	follow the app log
`granth_setup open` / `admin`	open the library / admin page
`granth_setup ports`	what is listening, and a warning if anything is reachable from other computers
`granth_setup ai`	change the AI (Ollama or a key) and restart
`granth_setup version` / `update` / `rollback`	versions, update from this package, go back
`granth_setup repair`	fix Python package conflicts
`granth_setup indic` / `indic remove`	add or remove the offline translator
`granth_setup uninstall [--all]`	remove the app

Routine

Daily (2 minutes) 1. `granth_setup status`: both ports answer, error count not rising. 2. Admin page > **Unanswered**: teach answers to repeated questions; **Reader corrections**; reader suggestions on **Translations**.

Weekly 1. Admin > **Visitors**: look at flagged addresses; label known ones; block abuse with a time limit. 2. Admin > **Log files**: recent warnings and errors. 3. Admin > **Audit trail** > **Check it has not been changed**. 4. Copy the daily audit files (`data/logs/audit/audit-*.jsonl`) to another computer or drive. A copy elsewhere is what makes the trail trustworthy.

Monthly 1. Back up (below) and test that the backup opens. 2. Admin > **System**: AI providers' daily use against their free limits. 3. Install updates if a new package is available.

Adding and changing books

- Copy files into the books folder, or upload on the admin page. A file whose content changed is read as a new book; remove the older copy on the Books tab.
- Large scanned books take time (OCR). Progress shows on the Books tab; two books are read at once (`INGEST_WORKERS`).
- To remove: admin > Books > **Remove**. Removing the file from the books folder does not remove it from the library.

Backup and restore

What to back up: the **data folder** (library database, index, audit files, settings copy), the **settings file** `app/.env` (contains keys: keep it private), and the **books folder**.

1. `granth_setup stop`
2. Copy the data folder, `app/.env` and the books folder to the backup drive.
3. `granth_setup start`

Restore on the same or a new computer: install the same version, `granth_setup stop`, copy the data folder and `.env` back into place, `granth_setup start`. Restoring only the database: copy a file from `data/backups/library-<date>.db` over `data/library.db` (and delete `library.db-wal` and `library.db-shm`) while the service is stopped.

Updating

1. Download the new package for your system and unzip it.
2. Run its installer. It stops the service, keeps the previous version for rollback, backs up the database, updates, repairs packages and starts again.
3. Check `granth_setup status`, ask one question, open the admin page.
4. Problem? `granth_setup rollback` returns to the previous version (settings, chats and index kept).

Changing the AI

`granth_setup ai`. For Gujarati and Hindi a **Gemini** key gives the best results. Several keys can be set at once in `.env`; they are used in turn when one reaches its free limit (`LLM_ORDER`). Check keys: `cd <ap`

`p folder> && ../venv/bin/python -m granth check-providers` (Windows: `..\venv\Scripts\python -m granth check-providers`).

Rotating the admin token

Stop the service, delete `data/admin_token`, start the service: a new token is created; read it from the same file. Everyone using the admin page must unlock again.

Moving to a new computer

Install on the new computer, stop it, copy the data folder, `.env` and books folder from the old one (see Backup and restore), start it, then uninstall from the old computer.

Incident: the library is down

1. `granth_setup status` and `granth_setup health`.
2. `granth_setup restart`.
3. Still down: `granth_setup logs`, and the end of `data/logs/service.err.log`. Match the message in TROUBLESHOOTING.md.
4. After an update: `granth_setup rollback`.
5. Database damaged: restore from `data/backups` (above).

Incident: abuse or unusual traffic

1. Admin > **Visitors**: sort out the flagged address; open it to see what it did.
2. **Block...** for 24 hours (or longer); the reason goes into the audit trail.
3. If the library is reachable from the internet, set `ACCESS_CODE` and lower `CHAT_RATE_PER_MIN`.

Incident: wrong or harmful answer reported

1. Find it: Admin > **Audit trail**, search a word from the question.
2. **Teach answer** (Unanswered) or approve the reader's correction, so the right answer is given from then on.
3. A wrong translation: correct it on **Translations** (it is remembered).

5. Monitoring and service tracking

Health checks

Check	How	Healthy
Library chat	`GET http://127.0.0.1:8000/healthz`	`{"ok": true, "version": "..."}` (checks the database too)
Admin page	`GET http://127.0.0.1:8001/healthz`	same
Both, as a command	`granth_setup health`	prints `OK ...`, exit code 0; `FAIL ...`, exit code 1
Everything an operator looks at	`granth_setup status`	service running, both ports answer, errors not rising
Details (admin token)	`GET http://127.0.0.1:8001/api/admin/status` with header `X-Admin-Token: <token>`	JSON: books, passages, AI providers (ready, cooling down, daily use), background load, audit writer queue
Audit trail intact	admin > Audit trail > *Check it has not been changed*; or in the app folder: `GRANTH_DATA=~/granth/data ../venv/bin/python -m granth audit-verify` (Windows: `set GRANTH_DATA=%LOCALAPPDATA%\Granth\data` then `..\venv\Scripts\python -m granth audit-verify`)	`database: OK`, `daily files: OK`

`/api/admin/status` → `load` shows `ai.active` and `ai.waiting` (AI requests running and queued), `worker_threads`, `translation_workers` and `threads_running`; `audit.waiting_to_write` should stay near 0.

Automatic checks

macOS / Ubuntu (cron) — check every 5 minutes, restart if down, keep a record:

```
crontab -e
*/5 * * * * ~/granth/granth_setup.sh health >/dev/null || (~/granth/granth_setup.sh restart; date >> ~/granth/data/logs/restarts.log)
```

Windows (Task Scheduler) — create a task that runs every 5 minutes:

```
schtasks /Create /SC MINUTE /MO 5 /TN "Granth health" /TR "cmd /c \"%LOCALAPPDATA%\Granth\granth_setup.cmd health || %LOCALAPPDATA%\Granth\granth_setup.cmd restart\""
```

The service itself already restarts after a crash (launchd *KeepAlive*, systemd *Restart=always*, Task Scheduler *restart every minute*); these checks also catch a hung process.

Uptime monitors (Uptime Kuma, Healthchecks.io agent, Zabbix, Nagios): an HTTP check on `http://127.0.0.1:8000/healthz` expecting `"ok":true`, run on the same computer.

Logs

File (in the data folder's `logs`)	Contents	Kept
`granth.log`	the app's own log (INFO, WARNING, ERROR)	rotated at 5 MB, 5 copies
`service.out.log`, `service.err.log`	what the service printed (start-up errors)	grows; trim when large
`audit/audit-YYYY-MM-DD.jsonl`	the audit trail, one file per day, hash-chained	`AUDIT_RETENTION_DAYS` (365)
`unanswered.jsonl`	questions the books could not answer	grows
`chat_history.txt`	readable transcript (if `TRANSCRIPT_LOG=1`)	grows

On Ubuntu with systemd also: `journalctl --user -u granth`. On Windows: Task Scheduler history of "Granth library". All log files can be viewed and downloaded on the admin page > **Log files**.

What to look for in `granth.log`:

Line	Meaning
`Granth x.y.z up: N books, M chunks, embeddings=..., llm=[...]`	a successful start
`provider X failed (HTTP 429 ...)`	a free AI key hit its limit; the next provider is used
`translation by X rejected (...)`	an AI translation failed the quality check and was retried
`offline translator unavailable: ...`	IndicTrans2 not usable; translation uses the AI
`ingest failed for book N`	a book could not be read (reason follows)
`audit write failed`	investigate: disk full or database locked

Activity and usage

- Admin > **Visitors**: who uses the library (by network address), how much, and flags for unusual use.
- Admin > **Audit trail**: every action, searchable and exportable as CSV for reports.
- Admin > **Learning**: questions in the last 7 days, answered, not found, rated helpful.
- Admin > **System**: each AI provider's use today against its daily limit.

Alerts worth acting on

Signal	Action
`health` fails twice in a row	`granth_setup restart`; read the log
Errors in `granth.log` rising	Log files tab; TROUBLESHOOTING.md
Many *Not found in the books*	add books, teach answers, approve learned words
A provider always *cooling down*	add another free key, or use Gemini

Signal	Action
Flagged addresses on Visitors	look, label or block
Disk space low (`status`)	trim `service.*.log`, remove old `data/backups`, remove unused Ollama models (`ollama rm`)
Audit check not intact	stop and investigate; compare with the copy kept elsewhere

6. Troubleshooting

Start with `granth_setup status` and `granth_setup logs` (`granth_setup` = `~/granth/granth_setup.sh` on macOS/Ubuntu, `%LOCALAPPDATA%\Granth\granth_setup.cmd` on Windows). The admin page > **Log files** shows recent warnings and errors.

Installation

Message or symptom	Cause	Fix
`Homebrew is required` (macOS)	Homebrew missing	install from https://brew.sh, run setup again
`Python 3.12 not found` (macOS)	brew could not install it	`brew install python@3.12`
`Python's venv module is missing` (Ubuntu)	`python3-venv` not installed	`sudo apt install python3-venv`
`Python 3.10 or newer is needed` (Ubuntu 20.04)	old Python	upgrade to Ubuntu 22.04+, or `PYTHON_BIN=/path/to/python3.12 ./granth_setup.sh`
Windows: "Windows protected your PC"	SmartScreen on a downloaded script	*More info* > *Run anyway*
Windows: `Python was installed but cannot be found yet`	PATH not refreshed	close the window, open a new one, run `granth_setup.cmd` again
Windows: runs `python` from the Microsoft Store	Store placeholder	setup ignores it; install Python from python.org or let setup use winget
`pip install failed`	no internet, proxy, or disk full	check internet and free space; run setup again
`ERROR: pip's dependency resolver ... conflicts`	packages from an older version	`granth_setup repair`
`the service did not answer within 90 s`	app failed to start	read the log lines shown; see "Service" below
`the app folder is missing next to this script` (Windows)	ran from inside the zip	extract the whole zip first

Service

Symptom	Fix
`NOT ANSWERING` on port 8000	`granth_setup restart`; then `granth_setup logs`
`address already in use` in the log	another program uses the port: `granth_setup ports`; choose others with `PORT` / `ADMIN_PORT` and run setup again
macOS: `launchd refused to start the service`	`launchctl bootout gui/$(id -u)/com.granth.app` then `granth_setup start`
macOS: books not read, `Operation not permitted`	books folder is inside Documents/Desktop/Downloads: move it (e.g. `~/GranthBooks`) or give Python *Full Disk Access*
Ubuntu: stops when you log out	`sudo loginctl enable-linger $USER`

Symptom	Fix
Ubuntu: `no systemd user session` warning	normal in containers/WSL: it runs as a background process; `granth_setup start` after a reboot if cron is not available
Windows: does not start after log-on	Task Scheduler > "Granth library" > History; or `granth_setup.cmd start`; corporate PCs may block tasks: setup then uses a Startup-folder shortcut
Windows: a console window flashes	update to the latest package (OCR steps run without windows)

Admin page

Symptom	Fix
"That admin token is not correct"	copy it again from `data/admin_token` (no spaces)
Admin page does not open from another device	by design: it answers only on the library computer
Admin page opens the chat instead	use port 8001 (`ADMIN_PORT`), not 8000

Books

Symptom	Cause	Fix
Book *failed*: "no text found"	a scan without OCR tools	install Tesseract and Poppler (setup does), then **Read again**
Gujarati scan reads as nonsense	OCR language missing	macOS: `brew install tesseract-lang`; Ubuntu: `sudo apt install tesseract-ocr-guj`; Windows: run setup again (downloads language data)
Book stays *queued*	another large book is being read	wait; two books are read at once (`INGEST_WORKERS`)
"Already in the library"	same file content	nothing to do

Answers

Symptom	Cause	Fix
Only passages, no written answer	no AI configured	`granth_setup ai`
"The AI is busy with other readers"	many questions at once	wait a moment; raise `LLM_CONCURRENCY` if the provider allows
Answers in poor Gujarati	small free model (e.g. Llama on Groq)	add a free **Gemini** key (`granth_setup ai`)
"Not found in the books" for something that is there	different words	rephrase; approve learned words on **Learning**; or **Teach answer**
Answers slow	free provider rate limits, or Ollama on a small computer	add a second key; use a smaller Ollama model

Translation

Message or symptom	Fix
"Please read and accept the translation terms first"	accept the terms in the pop-up (they changed)
"could not produce a reliable translation"	the AI's result failed the quality check; try later, or add a Gemini key
`Cannot access gated repo` / `401` (offline translator)	sign in to Hugging Face, accept the conditions on the three AI4Bharat model pages, then `granth_setup indic` and paste a Read token
`'NoneType' object has no attribute 'shape'` (offline translator)	newer `transformers` than AI4Bharat's model code supports: run `granth_setup indic` from version 2.11.0 or later (it installs 4.46; checked against their model code). Older packages (e.g. a `granth-2-19` folder) still have the bug
"The offline translator is still loading"	first start after a restart loads the model; wait a minute
Windows: offline translator install fails building a package	install *Microsoft C++ Build Tools* (`winget install Microsoft.VisualStudio.2022.BuildTools`), then `granth_setup.cmd indic`
Chrome offers to translate the page and the text changes	the app tells browsers not to; choose *Show original* once, or update to the latest version

Voice

Symptom	Fix
"No male voice available"	the voice service needs internet; or install a male system voice (Windows: Settings > Time & language > Speech; macOS: Accessibility > Spoken Content)
Microphone does nothing (Firefox)	Firefox has no voice input; use Chrome, Edge or Safari, or type
Phone/tablet does not read aloud automatically	the device allows sound only after a tap: tap **Read aloud**
"Microphone permission was denied"	allow the microphone for this site in the browser settings

Hands-free

Symptom	Fix
Stops listening after a moment of silence	update to 2.12.0 or later (it now keeps listening)
Does not listen again after an answer	update to 2.12.0 or later; check that *Read aloud* works on its own
"Tap the microphone to keep talking"	the browser needs a tap to listen again (Safari after the screen locked): tap the microphone
"Hands-free paused after a long silence"	press Hands-free again
Hears its own voice	use headphones, or lower the speaker volume
Firefox	no voice input in Firefox: use Chrome, Edge or Safari

Google Drive

Message	Fix
"folder not found, or not shared with the library"	share the Drive folder with the service-account e-mail shown on the Book sources tab, as Viewer
"Google did not accept the service-account key"	upload a fresh JSON key; the old one may have been deleted in Google Cloud
"Google refused access (403) … Drive API has not been used"	enable the **Google Drive API** in the same Google Cloud project
"Google Drive is not connected"	upload the service-account key, or save an API key (public folders only)
A folder from Google Drive for desktop is "not found"	Google Drive for desktop must be running; on a Mac choose *Mirror files*, not *Stream files*
Google Sheets / Slides not added	only Google Docs are read (exported as Word); save others as PDF in Drive

Phones and tablets

Symptom	Fix
Cannot open the library from a phone	by default only the library computer can; see *Reaching it from other devices* in INSTALL.md
Copy does not work over the local network	update to the latest version (copy works without HTTPS now)

Collecting information for support

`granth_setup status`, `granth_setup version`, the last 100 lines of `data/logs/granth.log` and `data/logs/service.err.log`, and the browser and device used. Remove API keys before sharing anything from `.env`.

7. Configuration

Settings live in the **settings file** `.env` in the app folder (`~/granth/app/.env`, Windows `%LOCALAPPDATA%\Granth\app\.env`). One `NAME=value` per line. Restart after a change (`granth_setup restart`). Setup manages the server and folder settings for you; the file also holds AI keys, so keep it private. `.env.example` in the package lists everything with comments.

AI providers (answers, summaries, translation)

Setting	Default	Meaning
`GEMINI_API_KEY`, `GROQ_API_KEY`, `CEREBRAS_API_KEY`, `NVIDIA_API_KEY`, `OPENROUTER_API_KEY`, `MISTRAL_API_KEY`, `DEEPSEEK_API_KEY`, `OPENAI_API_KEY`	empty	keys; any one is enough, several give automatic failover
`<NAME>_MODEL`, `<NAME>_BASE_URL`, `<NAME>_RPM`, `<NAME>_RPD`	built in	model, address, requests per minute and per day for a provider
`OLLAMA_MODEL`, `OLLAMA_URL`	empty, `http://127.0.0.1:11434`	AI on this computer
`CUSTOM_BASE_URL`, `CUSTOM_MODEL`, `CUSTOM_API_KEY`	empty	any OpenAI-compatible server
`LLM_ORDER`	`groq,gemini,cerebras,nvidia,openrouter,deepseek,openai,mistral,ollama,custom`	which provider is tried first for English
`TRANSLATE_ORDER`	`gemini,openai,mistral,deepseek,openrouter,cerebras,nvidia,groq,ollama,custom`	order for Indian languages (answers, summaries, translations)
`GEMINI_REASONING`	`none`	Gemini Flash's slow "thinking" step; `default` keeps Google's setting
`GLOBAL_LLM_DAILY_BUDGET`	`0` (no limit)	stop using the AI after this many requests a day
`CHAT_MODE`	`strict`	`strict` answers only from the books; `hybrid` also gives general answers (labelled) when the books have nothing

Translation

Setting	Default	Meaning
`TRANSLATE_ENGINE`	`auto`	`auto`: a strong AI key if set, else IndicTrans2 if installed; `llm`; `indictrans2`
`INDIC_PYTHON`	set by `granth_setup indic`	Python of the offline translator's own environment

Setting	Default	Meaning
`HF_TOKEN`	set by `granth_setup indic`	Hugging Face Read token for the IndicTrans2 models
`INDICTRANS_DEVICE`	`auto`	`cpu`, `mps` (Apple GPU), `cuda`; `auto` measures which is faster
`TRANSLATE_WORKERS`	`2`	book passages translated at the same time

Server and network

Setting	Default	Meaning
`HOST`, `PORT`	`127.0.0.1`, `8000`	library chat address and port
`ADMIN_HOST`, `ADMIN_PORT`	`127.0.0.1`, `8001`	admin page (`ADMIN_PORT=0` turns it off)
`ALLOW_LAN`	`0`	must be `1` to listen on anything other than this computer
`ACCESS_CODE`	empty	readers must enter this code before chatting
`ALLOWED_HOSTS`	empty	domain names allowed behind a reverse proxy
`TRUST_PROXY`	`0`	trust `X-Forwarded-For` from a local reverse proxy (for correct visitor addresses)
`CHAT_RATE_PER_MIN`, `MEDIA_RATE_PER_MIN`	`25`, `40`	per-address limits for questions and for voice/translation
`THREADS`	`64`	requests handled at the same time
`LLM_CONCURRENCY`	`8`	AI requests at the same time; others wait in a short queue

Books and search

Setting	Default	Meaning
`IMPORT_DIR`	books folder	folder watched for new books
`SCAN_INTERVAL`	`30`	seconds between folder checks
`INGEST_WORKERS`	`2`	books read at the same time
`DRIVE_SYNC_MINUTES`	`30`	how often connected Google Drive folders are checked (folders and keys are set on the admin page)
`MAX_UPLOAD_MB`, `MAX_PDF_PAGES`	`300`, `5000`	limits per book
`OCR_MODE`, `OCR_DPI`	`auto`, `200`	OCR for scans: `auto` (when a page has little or poor text), `force` (every page), `off`
`EMBED_BACKEND`, `EMBED_MODEL`	`auto`, `intfloat/multilingual-e5-small`	meaning-based search (`none` = keywords only)
`TOP_K`, `CHUNK_CHARS`, `CONTEXT_BUDGET`	`6`, `500`, `3800`	passages per answer, passage size, text given to the AI
`CACHE_TTL_HOURS`	`168`	reuse an identical recent answer

Voice

Setting	Default	Meaning
`TTS_BACKEND`	`auto`	male voices from the online voice service, cached; the browser's male voice as fallback
`STT_MODEL`	`small`	server speech recognition (optional `requirements-voice.txt`) for browsers without voice input

Learning, privacy and audit

Setting	Default	Meaning
`LEARN_FROM_FEEDBACK`	`1`	ratings re-rank passages; rephrasings teach words
`LEARN_MIN_READERS`, `LEARN_AUTO_APPROVE`	`2`, `3`	readers needed before a rating counts; before a learned word is added without review (`0` = admin only)
`AUDIT_RETENTION_DAYS`	`365`	keep audit records this long (`0` = forever); DPDP Rules 2025 ask for at least one year
`AUDIT_IP_MODE`	`full`	`full`, `truncate` (network only) or `hash` (a code per visitor)
`SESSION_RETENTION_DAYS`	`90`	delete chats older than this
`TRANSCRIPT_LOG`	`1`	readable transcript file
`APP_NAME`, `APP_MARK`	`Swaminarayan Sampradaya`, `ॐ`	default name and symbol shown to readers (the admin page System tab overrides them)
`LOG_LEVEL`	`INFO`	`DEBUG` for more detail

8. Browsers, phones and tablets

Supported browsers

Browser	Minimum	Chat, references, library, translation, downloads	Voice input (microphone)	Reading aloud (male voice)
Chrome / Edge (Windows, macOS, Linux, Android)	108	yes	yes	yes
Safari (macOS, iPhone, iPad)	15.4 (iOS/iPadOS 15.4)	yes	yes	yes; on iPhone/iPad after a tap
Firefox (Windows, macOS, Linux, Android)	115	yes	no (Firefox has no voice input: type, or install server speech recognition)	yes
Samsung Internet	21	yes	yes	yes
Older iPads (iPadOS before 15.4)	—	works with reduced styling (dialog fallback built in)	depends	after a tap

Reading aloud uses the library's own male voices (from the voice service, cached on the library computer). If that service cannot be reached, the browser's own **male** voice is used; if the device has none, nothing is read aloud and the reader is told (never a female voice).

Tested

What	How	Result
Phones: iPhone SE (320×568), iPhone 13, iPhone 15 Pro Max, Pixel 7, Galaxy S9+	Chromium with each device's screen size, pixel density, touch and user agent	no sideways scrolling; question box, menus, sidebar, Library and Settings dialogs fully on screen; buttons at least 38–44 px for fingers
Tablets: iPad Mini, iPad (7th gen), iPad Pro 11 landscape, Galaxy Tab S4	same	same
Desktop 1280–1920 px	Chromium	same
Admin page on all of the above	every tab	no sideways scrolling (wide tables scroll inside their box)
Full flows (ask, cite, read aloud, translate with terms, suggest, stop, download, admin tabs)	47 automated browser tests (jsdom) + 200+ server tests	all pass
Firefox and Safari engines	checked feature by feature against what those browsers support (fallbacks below)	not run in the real browsers in this build environment

Fallbacks built in

Feature	Newer browsers	Fallback
Full-height layout	`100dvh`	`100vh`
Highlight of selected books and current chat	`:has()`	a class set by the page
Colour mixing	`color-mix()`	fixed colours
Dialogs (Library, Settings, terms)	`<dialog>`	built-in stand-in for older Safari
Copy	Clipboard API (needs HTTPS or this computer)	classic copy, so it also works over plain HTTP on a local network
Hover-only buttons (chat rename, download, delete)	hover	always visible on touch screens
Sound on phones/tablets	autoplay after an answer	"tap Read aloud" message when the device blocks sound
Browser auto-translation (Chrome)	turned off for the app, each message marked with its real language	the app's own Translate button

Notes for phones and tablets

- By default only the library computer opens the library. To use phones and tablets, see *Reaching it from other devices* in INSTALL.md; HTTPS through a reverse proxy is best (the microphone needs HTTPS on phones).
- Add to Home Screen works on iPhone, iPad and Android (opens full screen with the library's colour).